

XPC Package Summary

A Universal Monte Carlo Shapley Toolkit

Generated technical summary

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Abstract

XPC is a compact Python package for Monte Carlo Shapley explanations over explicit feature groups. It supports ordinary tabular data and temporal tensors of shape (n, d, f) , where each sample-time location (n, d) can be explained while the model still receives the complete sequence context. The package intentionally exposes one explanation family: Monte Carlo Shapley over feature-group players. Kernel SHAP, permutation SHAP, GAM-specific logic, SMACH-specific logic, generative conditioning, and estimator-side known contribution handling are outside the estimator.

1 Design Goals

The package is organized around a small set of explicit abstractions:

`DataSpec`, `ModelAdapter`, `Masker`, `Conditioner`, `FeatureGroups`, `ShapleyExplainer`, `Explanation`.

Each abstraction owns one concern. Data layout is separate from model calling; model calling is separate from masking; masking is separate from conditioning; and signed Shapley estimation is separate from positive post-processing.

2 End-to-End Flow

3 Data Layout

3.1 Tabular Data

`DataSpec` represents independent rows:

$$X \in \mathbb{R}^{n \times f}.$$

The explained units are row indices $i \in \{1, \dots, n\}$. Model outputs are normalized to

$$\hat{y}(X) \in \mathbb{R}^{n \times H},$$

where $H = 1$ for scalar predictions and $H > 1$ for multi-output models. The resulting explanation has shape

$$\Phi \in \mathbb{R}^{n \times H \times G},$$

where G is the number of feature-group players.

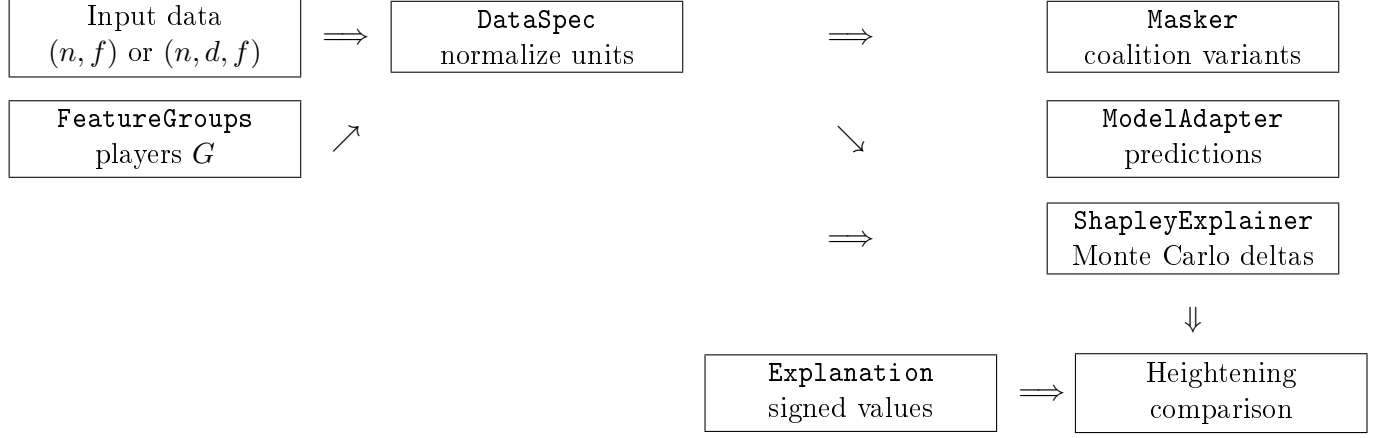


Figure 1: Main runtime flow. The estimator receives normalized data units, resolved feature-group players, masked coalition variants, and model predictions, then returns signed group Shapley values.

3.2 Temporal Tensor Data

`TimeSeriesTensorSpec` represents sequences:

$$X \in \mathbb{R}^{n \times d \times f}.$$

The explained units are pairs (i, t) . For a perturbed feature row at time t , XPC duplicates the complete sequence X_i , replaces only the selected time-location row, runs the sequence model, and extracts the output at time t . Model outputs are normalized to

$$\hat{y}(X) \in \mathbb{R}^{n \times d \times H},$$

and explanations have shape

$$\Phi \in \mathbb{R}^{n \times d \times H \times G}.$$

This lets temporal models use sequence context without forcing the user to flatten time into independent rows.

4 Model Adapters

`ModelAdapter` gives the explainer a uniform prediction boundary:

$$\text{predict}(X_{\text{batch}}) \rightarrow \text{numpy array}.$$

The package includes adapters for:

- NumPy callables and objects with a `predict` method;
- pandas DataFrame models through `PandasModelAdapter`;
- sklearn-style estimators through `SklearnModelAdapter`;
- PyTorch modules through `TorchModelAdapter`;
- external scripts through `ScriptModelAdapter`;
- R scripts through `RScriptModelAdapter`;
- custom registered classes through `register_model_adapter`.

Adapters do not change the Shapley algorithm. They only translate data and prediction outputs.

5 Feature Groups

Feature groups are the Shapley players. A group g is a disjoint subset of raw feature indices:

$$P_g \subseteq \{1, \dots, f\}.$$

The package supports:

- every feature as an individual player;
- custom named groups;
- remaining features as individual players;
- remaining features as one named group;
- always-present features.

Always-present features are included in every coalition and therefore influence the base value rather than receiving their own Shapley value.

6 Masking And Conditioning

For a unit feature row x and a present feature set S , a masker creates masked samples $\tilde{x}_S^{(m)}$ by preserving x_S and replacing missing features $\bar{S}$.

6.1 Baseline Masking

BaselineMasker replaces missing features with a deterministic baseline. Accepted baselines are:

zero, mean, median, scalar, feature vector, tensor, callable.

Tensor baselines may be full data-shaped tensors, which is especially useful for temporal explanations.

6.2 Random Empirical Masking

RandomMasker samples replacement rows from a reference/background matrix without conditioning:

$$\tilde{x}_{\bar{S}} \sim \hat{P}(X_{\bar{S}}).$$

6.3 Conditional Masking

ConditionalMasker is non-generative. It delegates to a conditioner that selects observed reference rows:

$$\tilde{x}_{\bar{S}} \sim \hat{P}(X_{\bar{S}} \mid X_S \approx x_S).$$

The included conditioners are:

- **EmpiricalConditioner**: nearest empirical neighbors in observed feature space;
- **GridConditioner**: empirical rows in matching grid cells, with nearest-neighbor fallback;
- **CallableConditioner**: user-defined empirical selection.

7 Monte Carlo Shapley Estimator

For each explained unit u , output horizon h , and feature-group player g , the Shapley value is

$$\phi_{u,h,g} = \mathbb{E}_{S \subseteq G \setminus \{g\}} [v_{u,h}(S \cup \{g\}) - v_{u,h}(S)],$$

where coalitions S are sampled from the Shapley subset distribution: first draw a coalition size uniformly from $0, \dots, G-1$, then draw that many players uniformly from $G \setminus \{g\}$.

The coalition value is an empirical model expectation:

$$v_{u,h}(S) = \frac{1}{M} \sum_{m=1}^M f_h(\tilde{x}_S^{(m)}),$$

where $M = \text{n_mask_samples}$. The estimator averages over n_coalitions sampled coalitions per player.

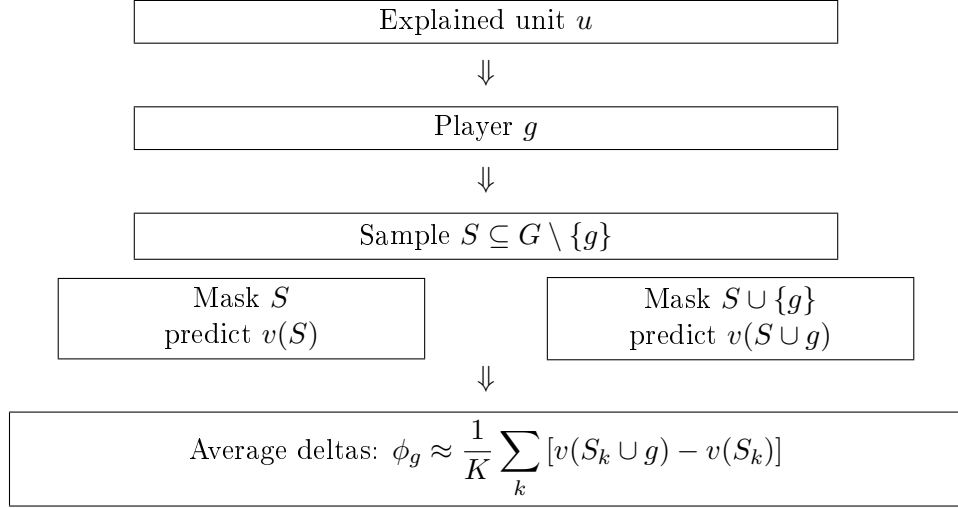


Figure 2: Estimator loop for one unit and one player.

8 Explanation Object

`Explanation` stores the signed estimator result:

$$\Phi, \quad b, \quad \hat{y},$$

where Φ has shape $\text{batch} + (H, G)$, b is the base value with shape $\text{batch} + (H)$, and $\hat{y}$ is the model prediction in the same output shape. The efficiency residual is

$$r = \hat{y} - \left(b + \sum_{g=1}^G \phi_g \right).$$

For deterministic baseline masking and additive models this can be exactly zero; for empirical and conditional Monte Carlo masking it is a diagnostic of sampling noise and the chosen conditioning distribution.

9 Precomputed Contributions

Known or externally supplied contributions are deliberately not used by `ShapleyExplainer`. Instead they enter through

`Explanation.from_contributions( $\Phi_{\text{ref}}, \dots$ ).`

This keeps the estimator mathematically clean while still allowing users to compare against synthetic ground truth, domain decompositions, or other explainers.

10 Heightening

Heightening is optional post-processing and is enabled by default. It preserves signed raw values, shifts them to non-negative values, normalizes to percentages, and allocates positive target totals:

$$\begin{aligned} o_{h,g} &= \alpha \max\left(-\min_u \Phi_{u,h,g}, 0\right), \quad \alpha \geq 0, \\ \Phi_{u,h,g}^+ &= \max(\Phi_{u,h,g} + o_{h,g}, 0), \\ p_{u,h,g} &= \frac{\Phi_{u,h,g}^+}{\sum_j \Phi_{u,h,j}^+}, \quad \text{part}_{u,h,g} = p_{u,h,g} \cdot T_{u,h}. \end{aligned}$$

The target $T_{u,h}$ defaults to the model prediction and must be non-negative. If all positive values are zero, XPC uses a uniform allocation over groups. Pass `heighten=False` to keep only signed values.

11 Reference Comparison

`Explanation.compare(reference)` computes aggregate comparison metrics: mean absolute error, root mean square error, bias, maximum absolute error, correlation, and per-group mean absolute error. References may be raw arrays or another `Explanation`; single-output batch vectors are broadcast to the package’s canonical output axis.

12 Minimal Usage

```
import numpy as np
from xpc import BaselineMasker, ShapleyExplainer

X = np.array([[1.0, 2.0], [3.0, 4.0]])
model = lambda x: x[:, 0] + 2.0 * x[:, 1]

exp = ShapleyExplainer(
    model,
    BaselineMasker("mean"),
    n_coalitions=128,
    random_state=0,
)(X)

print(exp.values.shape)      # (n, H, G)
print(exp.entropy_residual)
```

13 Practical Guidance

- Choose **FeatureGroups** according to the explanation question: groups are the players, not merely display labels.
- Use baseline masking for deterministic counterfactual reference values.
- Use random masking for unconditional empirical replacement.
- Use conditional masking when missing features should be replaced by observed rows compatible with the present coalition.
- Increase **n_coalitions** to reduce Shapley subset noise and **n_mask_samples** to reduce masking expectation noise.
- Inspect **efficiency_residual**; it is a useful sanity check for sampling quality and model-output shape handling.
- Keep signed **Explanation.values** for analysis. Heightened parts are attached by default for positive target decompositions and can be disabled with **heighten=False**.